

This document provides a guide to the files provided for replication of “Oligopolistic Price Leadership and Mergers: The United States Beer Industry.”

Changelog

- Added document describing an issue with the replicability of the consumer draws used in Miller and Weinberg (2017). The present paper relies on demand estimates generated in that paper.

Data Construction

The replication data we have provided differs from the IRI data used in the study, and is intended to help provide an understanding of the computer code that generates the results. In constructing the replication data, we a number of steps to reduce the overall computational burden and mask the IRI scanner data:

- Five of the 37 regions are included.
- Eight of the 39 products are included. This includes a 12-pack product for each of the five firms, and a 24/30-pack product for each of the domestic firms. The shares of the ABI and MillerCoors products are inflated to roughly preserve the firm-level market shares.
- A random perturbation is applied to all market shares and prices.

Although the IRI data that we use in the study are confidential, they are available for purchase from IRI. For more information see: <https://www.iriworldwide.com/en-us/solutions/academic-data-set>

Data Files

There are three data files in the “raw data” folder:

1. small_scanner.mat is a Matlab dataset that contains the prices, quantities, and attributes of the products. It differs from the IRI data used in the study, which is confidential. The data provided are intended to help provide an understanding of the computer code that generates the results. We a number of steps to reduce the overall computational burden and mask the underling IRI scanner data:
 - Five of the 37 regions are included.
 - Eight of the 39 products are included. This includes a 12-pack product for each of the five firms, and a 24/30-pack product for each of the domestic firms. The shares of the ABI and MillerCoors products are inflated to roughly preserve the firm-level market shares.
 - A random perturbation is applied to all market shares and prices.

In the file, columns correspond to variables and rows correspond to year-quarter-region-brand-size observations. The Matlab function `main_data.m` provides details on how to construct the variables used in project based on the scanner data.

2. small_demosE.mat is a Matlab dataset that contains the demographic draws for the five regions. It can be downloaded from the supplementary material provided for Miller and Weinberg (2017) on the Econometric Society website:

<https://www.econometricsociety.org/publications/econometrica/2017/11/01/understanding-price-effects-millercoors-joint-venture>

See the file “demos.csv.” It is available for download for anybody with a subscription to *Econometrica*. Rows are for year-region combinations and columns are 500 demeaned income draws. The file titled “Replication Note for ECMA.pdf” discusses the replicability of these draws.

3. small_dresgmm2.mat is a Matlab structure that contains the demand results published in Miller and Weinberg (2017, *Econometrica*). The coefficients can be verified by checking against that article. We take them as given here. The article can be downloaded from the Econometric Society using the link provided just above.

A number of intermediate files are created during replication. These are stored in the subfolder “analysis.” As we loop through different timing parameters, $\eta=(0.20,0.25,0.26,0.27,0.30,0.35,0.40)$, some of the intermediate files are slotted into subsubfolders (e.g., “df_200” for $\eta=0.20$). The precise contents of the intermediate file can be understood through the code, which we describe next.

Code

The replication can be accomplished by running `main_starter.m`, after adjusting the paths at the top of the program. No further action is required. The program relies on two libraries of functions written by the author, contained in the subfolders “functions” and “idcheck.” Each function contains comments that detail what the function does, and how it relates to the other functions. The replication function also relies on a package titled “mtimesx” that enables three-dimensional multiplication and thereby speeds up the calculation of demand derivatives. The subfolder “mtimesx” contains the needed files, and `main_starter.m` adds this subfolder to the path so Matlab understands the `mtimesx` command. More information on `mtimesx` can be found here:

<https://www.mathworks.com/matlabcentral/fileexchange/25977-mtimesx-fast-matrix-multiply-with-multi-dimensional-support>

Computation

Generating results using the replication code and data takes about two weeks using a single processor on a computer with the following specifications:

Intel(R) Xeon(R) CPU E5-2630 v3 @ 2.40GHz / 32 GB RAM.

Results

The results folder contains .pdf files that are used to construct the figures and .txt files that are used to construct the tables. Here is a mapping from the Figures and Tables to the files in the folder:

Table 1: reprinted from Miller and Weinberg (2017, *Econometrica*)

Table 2: created in results_baseanalysis.m

sumstats_pq.txt

Table 3: created in results_costregs.m

spec_df_coef_mod1.txt contains the coefficients.

spec_df_serr_mod1.txt contains the standard errors.

spec_df_mval_mod1.txt contains the other statistics.

Table 4: created in results_baseanalysis.m

eqeffects.txt

Table 5: created in results_baseanalysis.m

mean_markups.txt contains the number in the table.

markup_stats.txt contains additional statistics cited in the text.

Table 6: created in results_cfanalysis.m

cfmerger.txt

Table 7: created in results_cfanalysis.m

cfdecomp2.txt contains the statistics with two decimal points.

cfdecomp2_threedigits contains the statistics with three decimal points.

Table 8: created in results_cfanalysis.m

cfmergerwelfare.txt

Table G1: created in results_costregs.m

spec_df_coef_mod2.txt contains the coefficients.
spec_df_serr_mod2.txt contains the standard errors.
spec_df_mval_mod2.txt contains the other statistics.

Table G2: created in results_costregs.m

spec_df_coef_mod4.txt contains the coefficients.
spec_df_serr_mod4.txt contains the standard errors.
spec_df_mval_mod4.txt contains the other statistics.

Figure 1: reprinted from Miller and Weinberg (2017, *Econometrica*)

Figure 2: created in idgraph.m

digraph_a_2007.pdf, idgraph_b_2007.pdf

Figure 3: created in results_baseanalysis.m

prof_abi_07.pdf, prof_abi_10.pdf, prof_mill_07.pdf, prof_coor_07.pdf, prof_mc_10.pdf

Figure 4: created in results_baseanalysis.m

iccobs_07.pdf, iccobs_10.pdf

Figure 5: created in results_baseanalysis.m

smresw_abi_07.pdf, smresw_coors_07.pdf, smresw_abi_10, smresw_millercoors_10.pdf,
smresw_hhi_07.pdf, smresw_hhi_10.pdf

Figure 6: created in results_cfanalysis.m

nommc_07.pdf, nommc_10.pdf, mmc_profit_2007.pdf, mmc_profit_2010.pdf

Figure G1: created in results_costregs.m

sm_scatter_bigsmall_t.pdf, sm_scatter_2v1_t.pdf, mc_scatter_2v1_t.pdf

Figure G2: created in results_costregs.m

sm_07_scatter_pvnp_t.pdf, sm_10_scatter_pvnp_t.pdf, mc_scatter_pvnp_t.pdf

Figure G3: created in results_cfanalysis.m

mcleader_10.pdf

Figure G4: created in results_baseanalysis.m

Impliedfrictions.pdf

Numerical results in Appendix C3: idcheckout.m

idcheckout.txt

Other notes

- I certify that the authors of the manuscript have legitimate access to and permission to use the data in this manuscript.
- With the small replication dataset, total run time is between one and two weeks.
- All of the data provided for replication are publicly available. The full IRI data set used in the study is not publicly available but can be obtained from IRI under contract.

References

Miller, Nathan and Matthew Weinberg (2017). "Understanding the Price Effects of the MillerCoors Joint Venture." *Econometrica* 85(6): 1763-1791.